



US012414022B2

(12) **United States Patent**
Kodaypak et al.

(10) **Patent No.:** **US 12,414,022 B2**

(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **DYNAMIC DISCOVERY OF NETWORK FUNCTIONS IN MOBILITY INTERWORKING ACROSS RADIO ACCESS TECHNOLOGIES**

(71) Applicant: **AT&T Intellectual Property I, L.P.**,
Atlanta, GA (US)

(72) Inventors: **Rajendra Prasad Kodaypak**,
Sammamish, WA (US); **Vikram**
Seenappa, Sammamish, WA (US)

(73) Assignee: **AT&T Intellectual Property I, L.P.**,
Atlanta, GA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/738,733**

(22) Filed: **Jun. 10, 2024**

(65) **Prior Publication Data**

US 2024/0334284 A1 Oct. 3, 2024

Related U.S. Application Data

(60) Continuation of application No. 17/974,946, filed on
Oct. 27, 2022, now Pat. No. 12,041,506, which is a
(Continued)

(51) **Int. Cl.**
H04W 36/14 (2009.01)
H04W 8/24 (2009.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04W 36/14** (2013.01); **H04W 8/24**
(2013.01); **H04W 36/00222** (2023.05); **H04W**
36/144 (2023.05); **H04W 48/16** (2013.01)

(58) **Field of Classification Search**

CPC H04W 36/14; H04W 36/144; H04W
36/00222; H04W 8/24; H04W 48/16

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,235,226 B1 3/2019 Shekhar et al.
10,361,843 B1 7/2019 Suthar et al.

(Continued)

OTHER PUBLICATIONS

"5G Service-Based Architecture (SBA)", <https://medium.com/5g-nr/5g-service-based-architecture-sba-47900b0ded0a>, Oct. 20, 2018, 19.

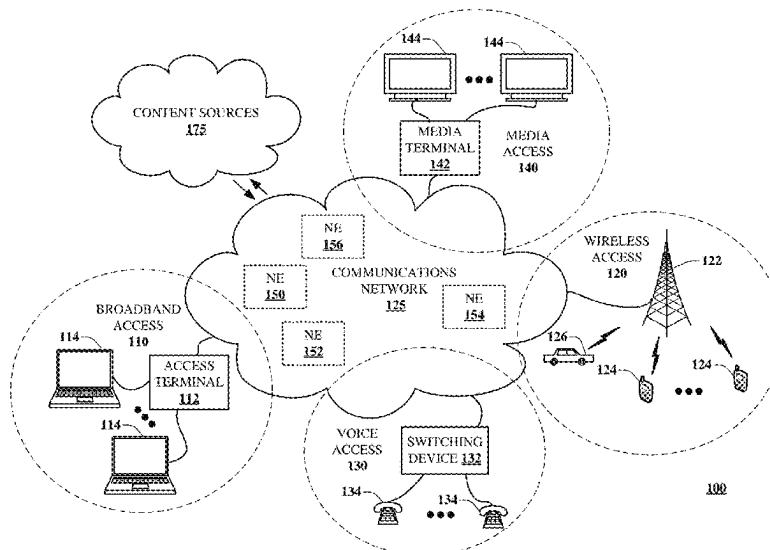
Primary Examiner — Dai Phuong

(74) *Attorney, Agent, or Firm* — Guntin & Gust, PLC;
Bryan S. Latham

(57) **ABSTRACT**

Aspects of the subject disclosure may include, for example, a machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations including receiving a first message from a mobile management entity (MME) in a Long-Term Evolution (LTE) network indicating a handover of user equipment (UE) to a fifth generation (5G) network covering a first area in which the UE is located; discovering an access mobility management function (AMF) in the 5G network responsive to the second message; receiving a second message from the AMF indicating a handover of the UE to the LTE network; and discovering a second mobile management entity (MME) in the LTE network covering a second area in which the UE is located. Other embodiments are disclosed.

20 Claims, 8 Drawing Sheets



Related U.S. Application Data

division of application No. 16/924,886, filed on Jul. 9, 2020, now Pat. No. 11,516,718.

(51) **Int. Cl.**

H04W 36/00 (2009.01)

H04W 48/16 (2009.01)

(56)

References Cited

U.S. PATENT DOCUMENTS

10,728,817	B1	7/2020	Kaki et al.	
11,671,816	B2	6/2023	Zong et al.	
12,127,249	B2 *	10/2024	Li	H04W 74/0808
2013/0040693	A1	2/2013	Chen et al.	
2018/0227737	A1	8/2018	Wu	
2018/0352092	A1	12/2018	Rajendran et al.	
2018/0367321	A1	12/2018	Stammers et al.	
2019/0394684	A1	12/2019	Li et al.	
2020/0007590	A1	1/2020	Dodd-Noble et al.	
2020/0120550	A1	4/2020	Narayanappa et al.	

2020/0162962	A1	5/2020	Li et al.	
2020/0245218	A1	7/2020	Wang et al.	
2020/0383010	A1	12/2020	Zhu et al.	
2020/0396587	A1	12/2020	Kim et al.	
2021/0029608	A1	1/2021	Dodd-Noble et al.	
2021/0176769	A1	6/2021	Chou et al.	
2021/0258766	A1	8/2021	Watanabe et al.	
2021/0306875	A1	9/2021	Zhang	
2021/0377838	A1	12/2021	Ali et al.	
2022/0014993	A1	1/2022	Kodaypak et al.	
2022/0060388	A1	2/2022	Li	
2022/0103443	A1	3/2022	Mary et al.	
2022/0150778	A1	5/2022	Jin et al.	
2022/0191764	A1 *	6/2022	Ni	H04W 36/322
2022/0377613	A1	11/2022	Wafa et al.	
2023/0007556	A1 *	1/2023	Qiao	H04W 36/322
2023/0048777	A1	2/2023	Kodaypak et al.	
2023/0189058	A1 *	6/2023	Yeh	H04W 48/18 370/329
2023/0276220	A1 *	8/2023	Wang	H04W 60/00 370/331

* cited by examiner

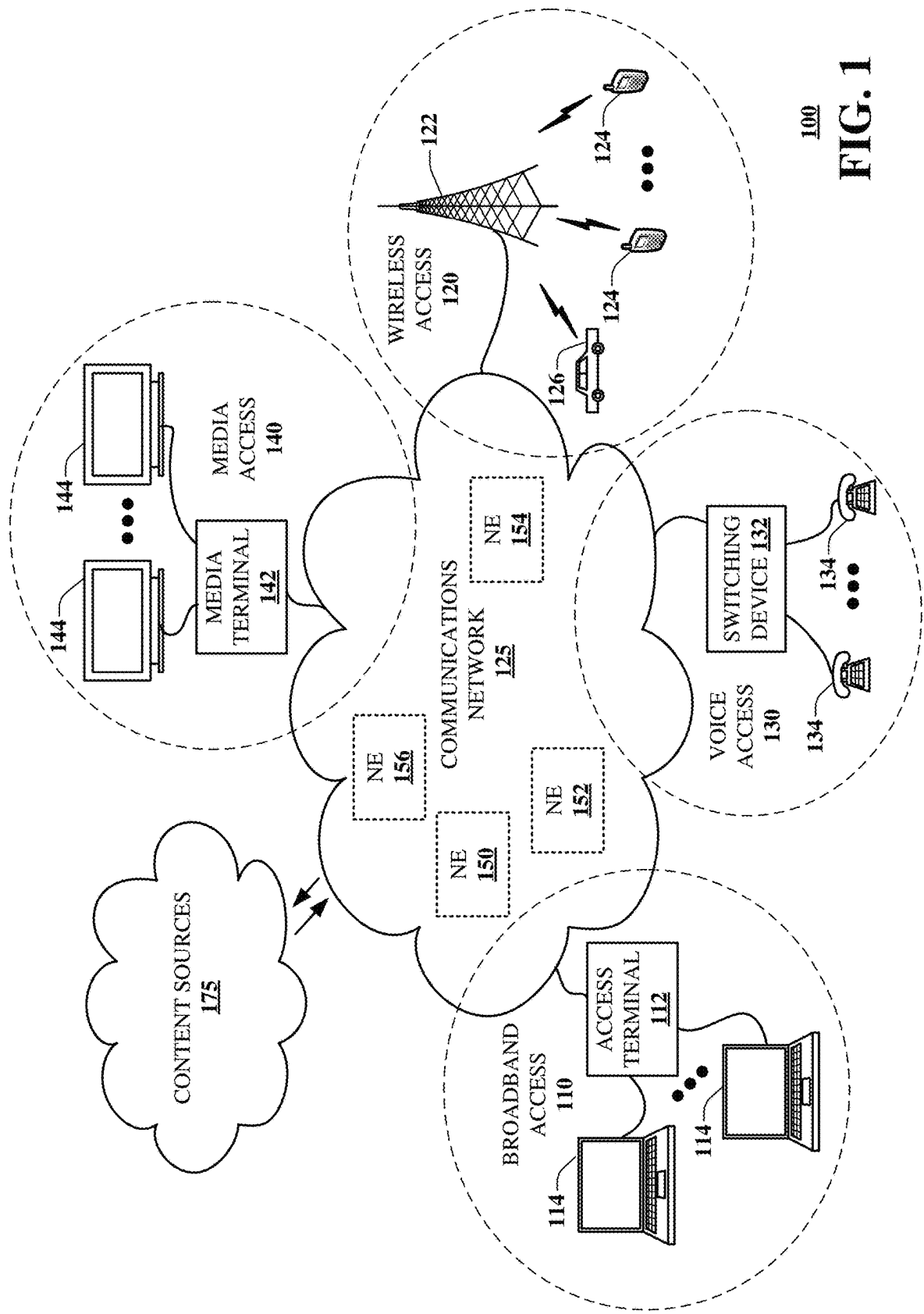
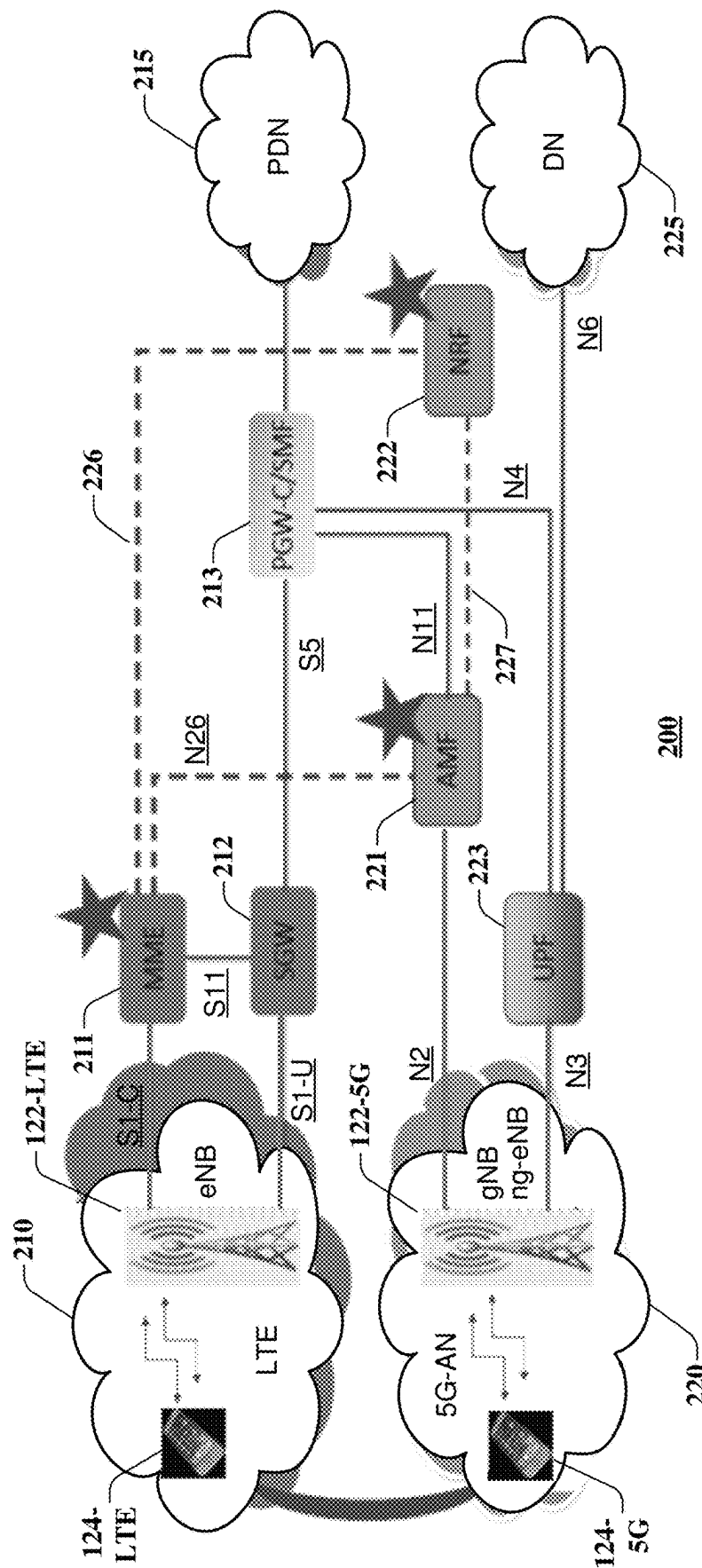


FIG. 1



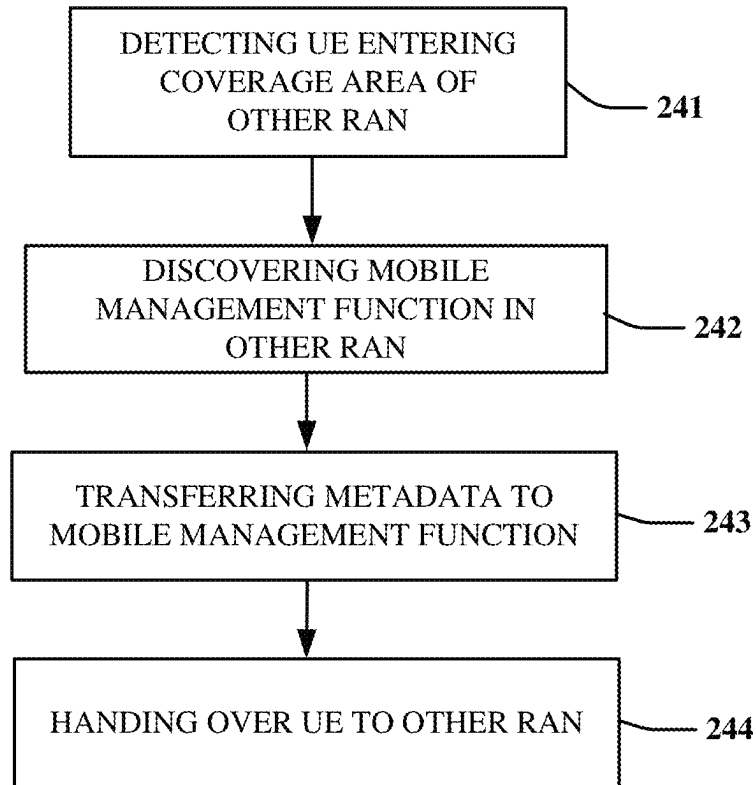
MME + AMF Metadata

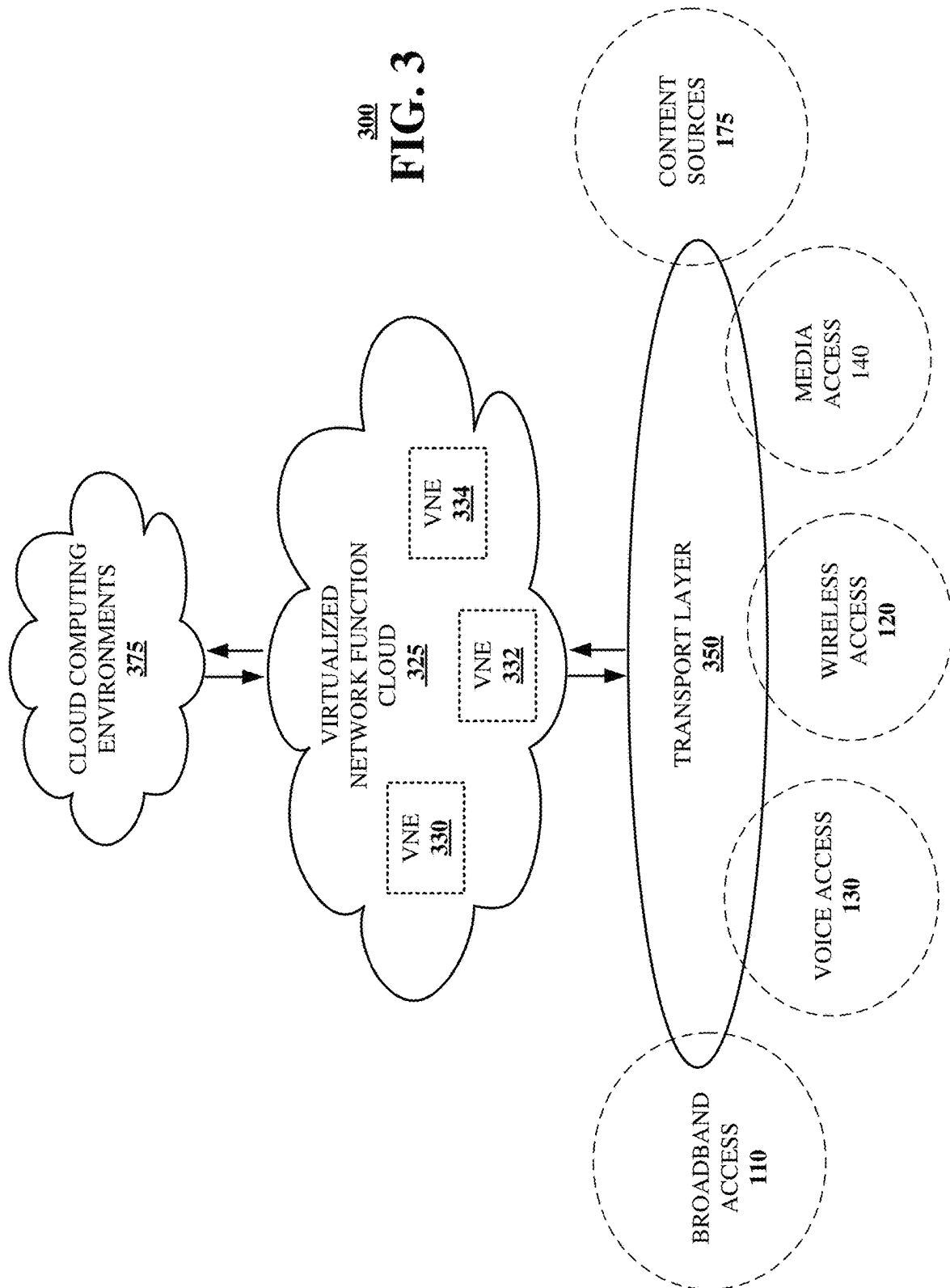
MME

- *MME FQDN*
- *MME Name*
- *MMEGI*
- *Status*
- *Heartbeat timer*
- *IPv4/IPv6Address*
- *LTE-TAList*
- *Capacity*
- *Priority*
- *Load*
- *Locality*
- *ServicesSupported*

AMF

- *Instance ID*
- *InstanceName*
- *Status*
- *Heartbeat timer*
- *IPv4/IPv6Address*
- *FQDN*
- *5GS-TAList*
- *Slices(NSSAI/NSI)*
- *Capacity*
- *Priority*
- *Load*
- *Locality*
- *ServicesSupported*

**FIG. 2C**



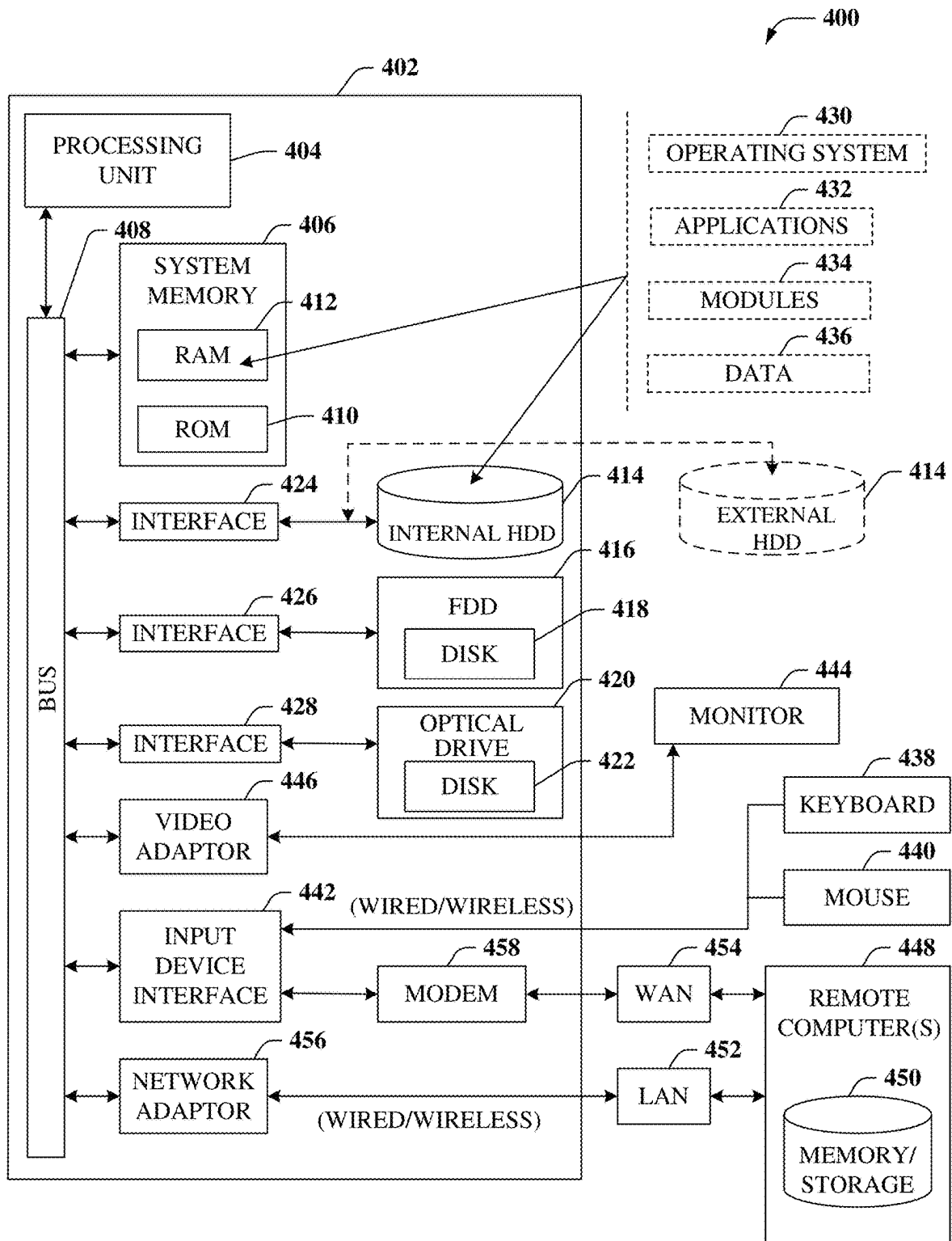


FIG. 4

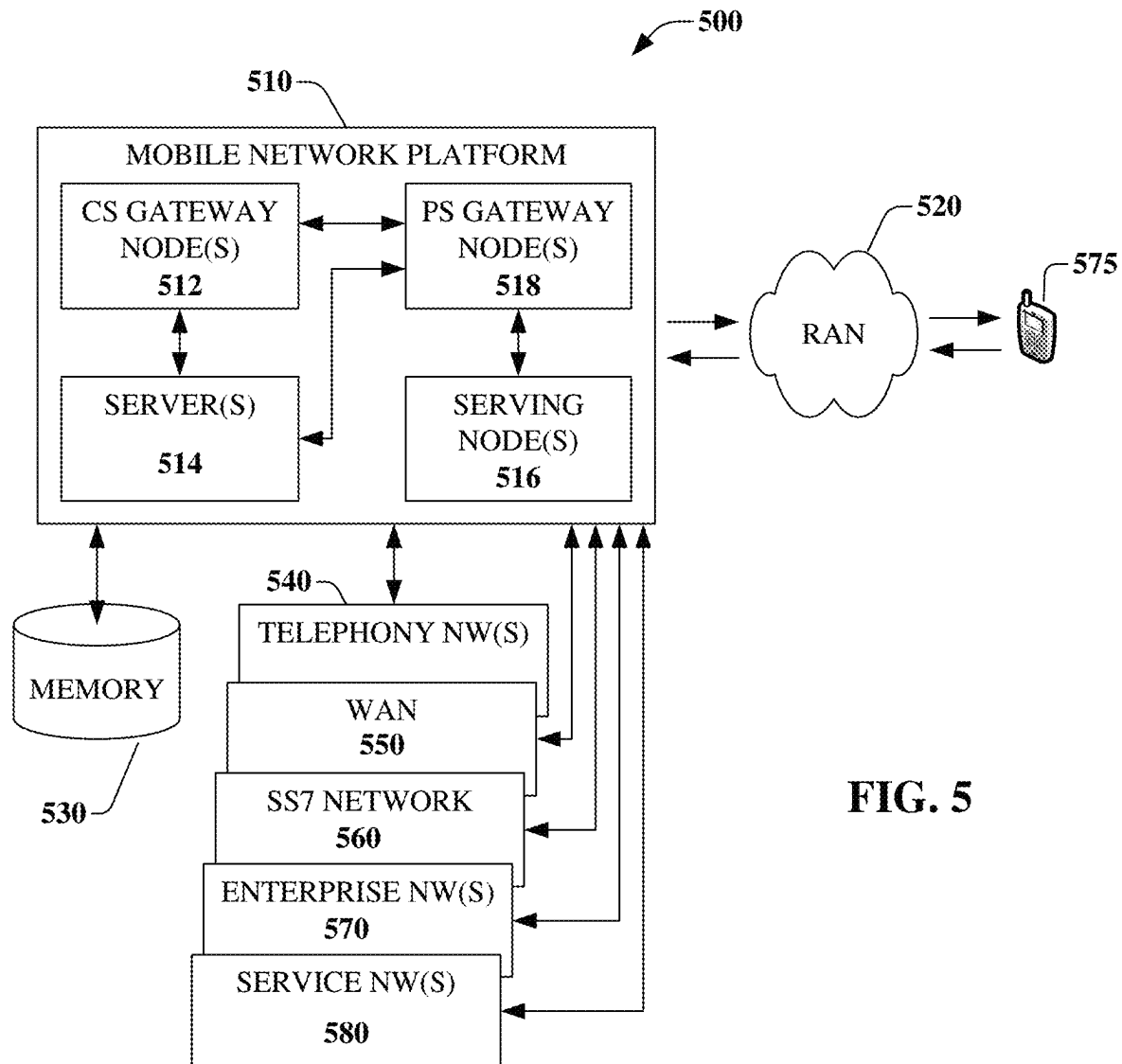
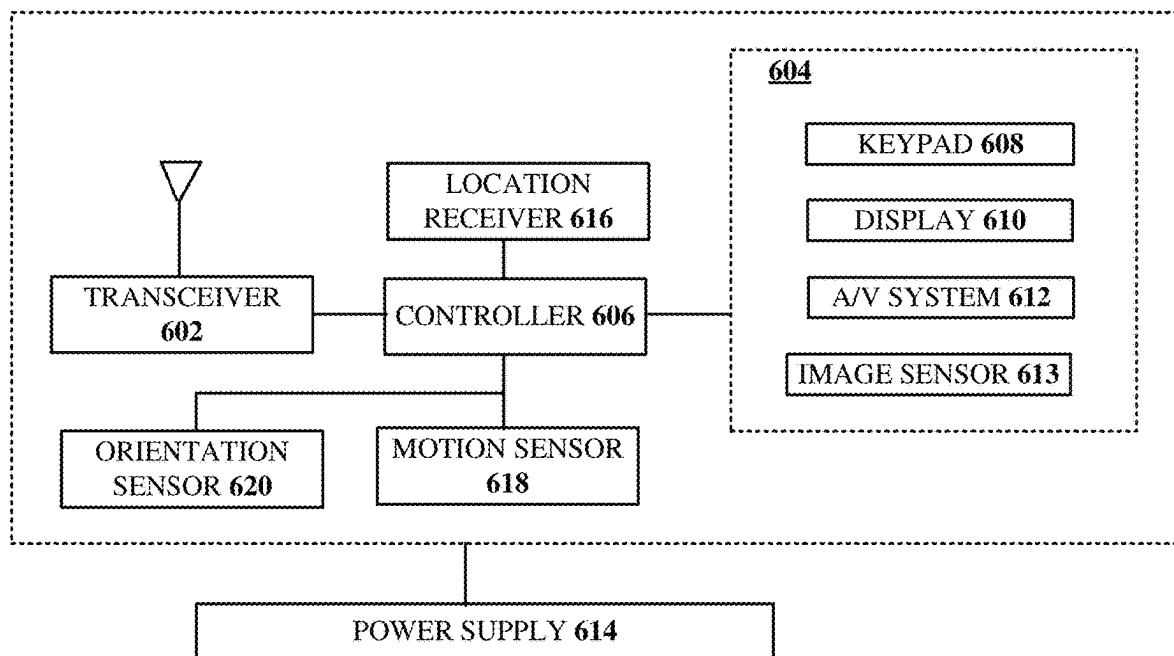


FIG. 5



600
FIG. 6

1

DYNAMIC DISCOVERY OF NETWORK FUNCTIONS IN MOBILITY INTERWORKING ACROSS RADIO ACCESS TECHNOLOGIES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/974,946 filed on Oct. 27, 2022, which is a divisional of U.S. patent application Ser. No. 16/924,886 filed on Jul. 9, 2020 (now U.S. Pat. No. 11,516,718). All sections of the aforementioned application are incorporated by reference herein in their entirety.

FIELD OF THE DISCLOSURE

The subject disclosure relates to dynamic discovery of network functions in mobility interworking across radio access technologies.

BACKGROUND

The fifth generation (5G) wireless technology evolution is a critical milestone in the telecom/IT industry that will disrupt the connectivity model in the society. The capabilities of 5G will enable new and exciting use cases spread across several consumer and enterprise industry verticals. 5G will continue to leverage the evolving fiber/metro ethernet transport and fourth generation (4G) Long Term Evolution (LTE) network infrastructure investments made by carriers to evolve into cloud native networks using service-based architectures. Hence 5G becomes a catalyst not only for mobility/telecom industry but to the entire society/world in a broader context.

BRIEF DESCRIPTION OF THE DRAWINGS

Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein: FIG. 1 is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.

FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a pair of communication networks in accordance with various aspects described herein.

FIG. 2B depicts a table illustrating a metadata sample shared between mobile management functions in two different radio access technologies.

FIG. 2C depicts an illustrative embodiment of a method in accordance with various aspects described herein.

FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

The subject disclosure describes, among other things, illustrative embodiments for interworking between 5G and

2

4G. The interworking in this disclosure applies to user equipment (UE) moving back and forth between 5G and LTE technologies as well as between mobile management entity (MME) and access mobility management function (AMF) geographic pool boundaries in Idle and Connected Modes with assorted services. The UE could also be operational in a unicast/multicast services mode when moving between the radio access technologies. Such a concept can be extended broadly to next generation wireless technologies beyond LTE and 5G. With next generation UE devices supporting multiple radio access technologies, the mobility between these radio technologies should be simplified with a direct service-based interface between the controller elements. Interworking becomes an extremely critical component of the eco-system mobility services delivery. With its advanced features and multi-mode capabilities, UE devices will be on the move to do different things dynamically that were not possible with previous generations of wireless technologies. Other embodiments are described in the subject disclosure.

One or more aspects of the subject disclosure include a method of detecting, by a processing system including a processor, user equipment (UE) in a Long-Term Evolution (LTE) network that has another radio technology network; determining, by the processing system, that the UE has crossed a threshold and entered a coverage area of the another radio technology network; discovering, by the processing system, an access mobility management function (AMF) in the another technology network; transferring, by the processing system, metadata associated with the UE to the AMF; and handing over, by the processing system, cellular services to the another technology network.

One or more aspects of the subject disclosure include a device including: a processing system including a processor; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations including detecting user equipment (UE) in a fifth generation (5G) network; determining, by the processing system, that the UE is leaving a coverage area of the 5G network; discovering, by the processing system, a mobile management entity (MME) in a Long-Term Evolution (LTE) network covering an area that the UE is within; transferring, by the processing system, metadata associated with the UE to the MME; and handing over, by the processing system, cellular services to the LTE network.

One or more aspects of the subject disclosure include a machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations including receiving a first message from a mobile management entity (MME) in a Long-Term Evolution (LTE) network indicating a handover of user equipment (UE) to a fifth generation (5G) network covering a first area in which the UE is located; discovering an access mobility management function (AMF) in the 5G network responsive to the second message; receiving a second message from the AMF indicating a handover of the UE to the LTE network; and discovering a second mobile management entity (MME) in the LTE network covering a second area in which the UE is located.

Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system 100 in accordance with various aspects described herein. For example, system 100 can facilitate in whole or in part an LTE or 5G network, UEs, RAN nodes or various network functions. In particular, a communications network 125 is presented for providing broadband access 110 to a plurality of data terminals 114 via access terminal 112,

3

wireless access **120** to a plurality of mobile devices **124** and vehicle **126** via base station or access point **122**, voice access **130** to a plurality of telephony devices **134**, via switching device **132** and/or media access **140** to a plurality of audio/video display devices **144** via media terminal **142**. In addition, communication network **125** is coupled to one or more content sources **175** of audio, video, graphics, text and/or other media. While broadband access **110**, wireless access **120**, voice access **130** and media access **140** are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices **124** can receive media content via media terminal **142**, data terminal **114** can be provided voice access via switching device **132**, and so on).

The communications network **125** includes a plurality of network elements (NE) **150**, **152**, **154**, **156**, etc. for facilitating the broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or the distribution of content from content sources **175**. The communications network **125** can include a circuit switched or packet switched network, a voice over Internet protocol (VOIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, Ultra Wideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

In various embodiments, the access terminal **112** can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals **114** can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

In various embodiments, the base station or access point **122** can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices **124** can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

In various embodiments, the switching device **132** can include a private branch exchange or central office switch, a media services gateway, VOIP gateway or other gateway device and/or other switching device. The telephony devices **134** can include traditional telephones (with or without a terminal adapter), VOIP telephones and/or other telephony devices.

In various embodiments, the media terminal **142** can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal **142**. The display devices **144** can include televisions with or without a set top box, personal computers and/or other display devices.

In various embodiments, the content sources **175** include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

In various embodiments, the communications network **125** can include wired, optical and/or wireless links and the network elements **150**, **152**, **154**, **156**, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs,

4

servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

FIG. **2A** is a block diagram illustrating an example, non-limiting embodiment of a pair of communication networks in accordance with various aspects described herein. The first network of the pair comprises an LTE network **210**. The LTE network **210** includes a base station (eNB) **122-LTE**, a Mobile Management Entity (MME) **211**, a Serving Gateway (SGW) **212**, a Packet Data Network Gateway Control/Session Management Function (PGW-C/SMF) **213**, which provides access to a Packet Data Network (PDN) **215**. Standard interfaces between these components illustrated in FIG. **2A** include S1-C, S1-U, S5 and S11. The S1 user plane external interface S1-U is between the eNB **122-LTE** and the SGW **212**. The S1 control plane interface S1-C is between the eNB **122-LTE** and the MME **211**. The S5 interface provides user plane tunneling and tunnel management between the SGW **212** and the PGW-C/SMF **213**. The S11 interface is between the MME **211** and SGW **212**.

The second network comprises a 5G Access Network (5G-AN) **220**. The 5G-AN **220** includes a base station (gNB) **122—5G**, an Access Mobility Management Function (AMF) **221**, a Network Repository Function (NRF) **222** and a User Data Plane Function (UPF) **223** which provides access to a data network (DN) **225**. The AMF **221** performs key mobility management functions including connection management, registration management, location services management, fallback for voice and emergency services when supporting mobility between 5G and LTE and/or Wi-Fi radio access technologies. AMF **221** also supports short message service (SMS) messaging, Communication Assistance for Law Enforcement Act (CALEA) lawful interception, and wireless emergency alerts that are critical to meet the regulatory requirements. The AMF **221** is distinct from other controller network functions, as it is the brain behind the 5G network and a critical control plane network function that makes interworking with MME in LTE technology possible.

The PGW-C/SMF **213** can also be considered to be part of the 5G-AN **220**. Standard interfaces between these components illustrated in FIG. **2A** include user plane data traffic interfaces N3 and N6, which provide packet processing and traffic aggregation closer to the network edge, increasing bandwidth efficiencies while reducing network size. The N3 interface is between the gNB **122—5G** and the UPF **223**. The N6 interface is between the UPF **223** and the DN **225**. Additionally, standard signaling/control plane interfaces illustrated in FIG. **2A** are N2, N11 and N4. Network functions in 5G control plane interact using these service-based interfaces. The N2 interface is between the gNB **122—5G** and the AMF **221**. The N11 interface is between AMF **221** and the PGW-C/SMF **213**. The N4 interface is between the UPF **223** and the PGW-C/SMF **213**. There are a number of additional standard interfaces (not illustrated) that provide other services-based network functions (also not illustrated).

Seamless interworking between 4G/LTE and 5G technologies is critical for the early deployment of 5G Option 3x Non-Stand Alone (NSA) Systems. In the current LTE deployments that are already mature and evolving to support 5G technologies, the Domain Name Service (DNS) systems are facing various shortcomings. This is due to the fact that the DNS records are becoming extremely complex to serve

combinations of Third Generation Partnership Project (3GPP) standards defined service types, device usage types, core network functions (gateways with multiple service and vendor flavors), applications and services. DNS based methods are complex and not scalable in future. Interworking between 4G/LTE and 5G technologies adds to the complexity of this DNS design. Interworking between 5G and 4G technologies involves the discovery of two key core network functions—MME 211 in LTE and the AMF 221 in 5G, by the respective network function in the other respective technology. The DNS based discovery model proposed by 3GPP standards is an inefficient, complex network design model. DNS resolves the Internet Protocol (IP) address of the respective node given a domain name. A single DNS node poses multiple challenges in terms of records creation for performing IP address resolution (often manually provisioned and prone to human errors leading to network outage in certain scenarios), with the addition of new standards defined service types, maintenance of legacy and new service types supporting end user services (data, mobile edge computing (MEC), network edge computing (NEC), home public land mobile network (PLMN), roaming, 5G, etc.), hardware or software upgrades, sync up across the DNS regional clusters and undesirable capital and operational expenses (CapEx/OpEx) due to legacy network infrastructure.

Seamless interworking is critical for the early deployment and adoption of 5G Option 2 Stand Alone (SA) deployments as well, so that they can work efficiently during mobility. With the legacy DNS based design, the MME 211 and the AMF 221 network functions discovery process tends to become cumbersome due to the addition of multiple IP address resolution records with new service types. This problem gets exacerbated in a geometric progressive fashion due to the considerable number of network functions deployed in these networks, and new deployments in the next 5 years.

In an embodiment, a new direct Hypertext Transfer Protocol (HTTP)/2 Representational State Transfer Application Programming Interface (REST API) triggering model 226 between the MME 211 in LTE and the NRF 222 in the 5G-AN 220 is used to discover the AMF 221 function when user equipment (UE) 124-LTE is transferred from the LTE network 210 to the 5G-AN 220 without using a DNS protocol. The MME 211 uses this REST API 226 to discover the AMF 221 instance or set of instances based on service-based architecture (SBA) interactions with the NRF 222 as a single repository network function that maintains the mapping of AMF 221 geographic pools. The NRF 222 not only maintains AMF 221 and MME 211 metadata but also other key network functions and attributes such as LTE Tracking Areas, 5G Tracking Areas that are critical for AMF 221 to handle other mobility services. Hence the discovery of right AMF 221 instance by the MME 211 via a direct SBA interface is much more effective in transitioning the UE from 5G to LTE. Based on a hierarchical deployment mode, the MMEs 211 in any geographic pool region will be able to discover an AMF 221 via the NRF 222 with proper mapping to ensure there is seamless service continuity across the mobility network domains.

In the opposite direction, a new direct REST API triggering model 227 through an SBA interface between the AMF 221 and NRF 222 discovers the MME 211 in LTE network 210 when UE 124—5G moves to the LTE network 210, without relying on a DNS protocol. Such enhanced service-based triggering, discovery and blacklisting of MME/AMF nodes during mobility is critical for simplified mobility core

network architecture design and evolution. This model enables new network capabilities such as core network slicing, simplify the network functions discovery during mobility across slices supporting multiple radio access technologies. The AMF 221 interacts through an SBA interface with the NRF 222 to discover the MME 211 in a comparable manner as described above during UE mobility between LTE and 5G. With the NRF 222 having all the mapping information of MME 211, LTE Type Allocation Code (TAC), 5G TAC, MME-LTE TAC, the AMF 221 can discover the MME 211 information to transfer the context and complete the session transfer into 5G seamlessly.

Additionally, an enhanced N26 interface, which is an inter-Core Network interface between the in the MME 211 and the AMF 221 will enable interworking between the LTE EPC and the 5G core. The enhanced N26 interface supports the functionalities of the LTE S10 interface (not shown, which is between different LTE MMEs) that are required for interworking with the 5G core. Messages sent on the enhanced N26 interface include:

- Context Request, Context Response & Context Acknowledge—are used to discover the peer node functions and are enhanced to support the discovery function.
- Forward Relocation Request & Forward Relocation Response—come into play in an NSA mode.
- Identification Request & Identification Response—checking to ensure that the exchange is for an authorized user transferring between networks technologies.
- Echo Request & Echo Response—used to maintain the health of the N26 link.
- Forward Access Context Notification & Forward Access Context Acknowledge
- Forward Relocation Complete Notification & Forward Relocation Complete Acknowledge
- Relocation Cancel Request and Relocation Cancel Response

These messages are transferred between the MME 211 and the AMF 221 based on auto discovery of the respective component via the NRF 222 using a REST API described above during mobility between LTE-5G and 5G-LTE. This method leverages direct interactions via HTTP/2 REST API triggers between the MME 211 and the NRF 222, as well as between the AMF 221 and the NRF 222. This will help simplify 5G-AN 220 design and implementation that is targeted to serve millions of users/devices across various industry verticals in the next several years.

The advantages of this approach are: 1. Simplified mobility core network architecture design with direct REST API triggering model between MME 211 and NRF 222 as well as AMF 221 and NRF 222 for peer node auto discovery during mobility. 2. Deliver seamless, direct and efficient interworking between 5G and LTE technologies. 3. Minimize dependencies on legacy DNS systems by transforming the network operational design and rely on enhanced services-based architecture framework towards smoother evolution. 4. Reduced CapEx/OpEx by eliminating additional functionality to be designed, implemented, operated and maintained in DNS. 5. Gradually migrate the DNS functions into a consolidated root NRF 222 network function in the 5G Core network. 6. Graceful evolution of root NRF 222 based core network topology design in cloud native environment. 7. Easily extendable to serve mobility between 5G and Wi-Fi and vice versa in case of LTE coverage holes.

FIG. 2B depicts a table 230 illustrating sample metadata stored on each of the MME 211 and the AMF 221. The metadata is used to exchange information between the MME 211 and the AMF 221 based on auto discovery of the

respective component by the NRF **222**. This metadata is not necessarily restricted to the information indicated. For example, other critical attributes can be transferred from MME **211** or AMF **221** to the NRF **222** for the NRF **222** to maintain a mapped set shared across the regions using a hierarchical NRF distributed model. Additional examples for IoT devices could include their sleep modes, device categories, extended timers, etc. This is extremely critical for disaster/failover situations in any given region impacting a specific cloud data center location serving mobility subscribers/devices.

FIG. **2C** depicts an illustrative embodiment of a method in accordance with various aspects described herein. As shown in FIG. **2C**, the method **240** begins in step **241** the system detects user equipment (UE) in either a Long-Term Evolution (LTE) network or a fifth generation (5G) network has crossed a threshold and is entering a coverage area of the other radio access network (RAN). Next, in step **242**, the system discovers the mobile management function of the other RAN. For example, an MME in an LTE network sends a message to an NRF that looks up the AMF in the 5G network, or vice-versa, depending on which network is currently providing cellular services to the UE.

Next, in step **243**, the mobile management function of the current RAN transfers metadata describing the UE to the mobile management function of the other RAN. For example, the MME in the LTE network sends the metadata to the AMF discovered in the previous step through a REST API.

Finally, in step **244**, the system transfers the UE to the other RAN, which provides cellular services.

While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. **2C**, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

Referring now to FIG. **3**, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, the subsystems and functions of system **200**, and method **240** presented in FIGS. **1**, **2A**, **2B**, **2C**, and **3**. For example, virtualized communication network **300** can facilitate in whole or in part an LTE or 5G network, UEs, RAN nodes or various network functions that can be virtualized.

In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

In contrast to traditional network elements—which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc.

For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) protocols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general-purpose processors or general-purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

As an example, a traditional network element **150** (shown in FIG. **1**), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it is elastic: so, the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized and might require special DSP code and analog front ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name service (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements do not typically need to forward substantial amounts of traffic, their workload can be distributed across a number of servers—each of which adds a portion of the capability, and which creates an overall elastic function with higher availability than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

The cloud computing environments **375** can interface with the virtualized network function cloud **325** via APIs that expose functional capabilities of the VNEs **330**, **332**, **334**, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud **325**. In particular, network workloads may have applications distributed across the virtualized network function cloud **325** and cloud computing environment **375** and in the commercial cloud or might simply orchestrate workloads supported entirely in NFV infrastructure from these third-party locations.

FIG. **4** illustrates a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. **4** and the following discussion are intended to provide a brief, general description of a suitable computing environment **400** in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment **400** can be used in the implementation of network elements **150**, **152**, **154**, **156**, access terminal **112**, base station or access point **122**, switching device **132**, media terminal **142**, and/or VNEs **330**, **332**, **334**, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment **400** can facilitate in whole or in part an LTE or 5G network, UEs or RAN nodes.

Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

The illustrated embodiments of the embodiments herein can also be practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

With reference again to FIG. **4**, the example environment can comprise a computer **402**, the computer **402** comprising a processing unit **404**, a system memory **406** and a system bus **408**. The system bus **408** couples system components including, but not limited to, the system memory **406** to the processing unit **404**. The processing unit **404** can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit **404**.

The system bus **408** can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory **406** comprises ROM **410** and RAM **412**. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer **402**, such as during startup. The RAM **412** can also comprise a high-speed RAM such as static RAM for caching data.

The computer **402** further comprises an internal hard disk drive (HDD) **414** (e.g., EIDE, SATA), which internal HDD **414** can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) **416**, (e.g., to read from or write to a removable diskette **418**) and an optical disk drive **420**, (e.g., reading a CD-ROM disk **422** or, to read from or write to other high capacity optical media such as the DVD). The HDD **414**, magnetic FDD **416** and optical disk drive **420** can be connected to the system bus **408** by a hard disk drive interface **424**, a magnetic disk drive interface **426** and an optical drive interface **428**, respectively. The hard disk drive interface **424** for external

drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer 402, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

A number of program modules can be stored in the drives and RAM 412, comprising an operating system 430, one or more application programs 432, other program modules 434 and program data 436. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM 412. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

A user can enter commands and information into the computer 402 through one or more wired/wireless input devices, e.g., a keyboard 438 and a pointing device, such as a mouse 440. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit 404 through an input device interface 442 that can be coupled to the system bus 408, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

A monitor 444 or other type of display device can also be connected to the system bus 408 via an interface, such as a video adapter 446. It will also be appreciated that in alternative embodiments, a monitor 444 can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer 402 via any communication means, including via the Internet and cloud-based networks. In addition to the monitor 444, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

The computer 402 can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) 448. The remote computer(s) 448 can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer 402, although, for purposes of brevity, only a remote memory/storage device 450 is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) 452 and/or larger networks, e.g., a wide area network (WAN) 454. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate

enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

When used in a LAN networking environment, the computer 402 can be connected to the LAN 452 through a wired and/or wireless communication network interface or adapter 456. The adapter 456 can facilitate wired or wireless communication to the LAN 452, which can also comprise a wireless AP disposed thereon for communicating with the adapter 456.

When used in a WAN networking environment, the computer 402 can comprise a modem 458 or can be connected to a communications server on the WAN 454 or has other means for establishing communications over the WAN 454, such as by way of the Internet. The modem 458, which can be internal or external and a wired or wireless device, can be connected to the system bus 408 via the input device interface 442. In a networked environment, program modules depicted relative to the computer 402 or portions thereof, can be stored in the remote memory/storage device 450. It will be appreciated that the network connections shown are examples and other means of establishing a communications link between the computers can be used.

The computer 402 can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

Turning now to FIG. 5, an embodiment 500 of a mobile network platform 510 is shown that is an example of network elements 150, 152, 154, 156, and/or VNEs 330, 332, 334, etc. For example, platform 510 can facilitate in whole or in part UEs. In one or more embodiments, the mobile network platform 510 can generate and receive signals transmitted and received by base stations or access points such as base station or access point 122. Generally, mobile network platform 510 can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, which facilitate both packet-switched (PS) (e.g., internet protocol (IP), frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform 510 can be included in telecommunications carrier networks and can be considered carrier-side components as discussed elsewhere herein. Mobile

network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology(ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510** can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform **510**. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) **518** for authorization/authentication and initiation of a data session, and to serving node(s) **516** for communication thereafter. In

addition to application server, server(s) **514** can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform **510** to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) **512** and PS gateway node(s) **518** can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN **550** or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform **510** (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. 1(s) that enhance wireless service coverage by providing more network coverage.

It is to be noted that server(s) **514** can comprise one or more processors configured to confer at least in part the functionality of mobile network platform **510**. To that end, the one or more processors can execute code instructions stored in memory **530**, for example. It should be appreciated that server(s) **514** can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

In example embodiment **500**, memory **530** can store information related to operation of mobile network platform **510**. Other operational information can comprise provisioning information of mobile devices served through mobile network platform **510**, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory **530** can also store information from at least one of telephony network(s) **540**, WAN **550**, SS7 network **560**, or enterprise network(s) **570**. In an aspect, memory **530** can be, for example, accessed as part of a data store component or as a remotely connected memory store.

In order to provide a context for the various aspects of the disclosed subject matter, FIG. 5, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

Turning now to FIG. 6, an illustrative embodiment of a communication device **600** is shown. The communication device **600** can serve as an illustrative embodiment of devices such as data terminals **114**, mobile devices **124**, vehicle **126**, display devices **144** or other client devices for communication via either communications network **125**. For example, computing device **600** can facilitate in whole or in part elements of an LTE or 5G network, including UEs, RAN nodes or various network functions.

The communication device **600** can comprise a wireline and/or wireless transceiver **602** (herein transceiver **602**), a

15

user interface (UI) **604**, a power supply **614**, a location receiver **616**, a motion sensor **618**, an orientation sensor **620**, and a controller **606** for managing operations thereof. The transceiver **602** can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, Wi-Fi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/10 GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver **602** can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VOIP, etc.), and combinations thereof.

The UI **604** can include a depressible or touch-sensitive keypad **608** with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device **600**. The keypad **608** can be an integral part of a housing assembly of the communication device **600** or an independent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high-volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals from an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power

16

sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, Wi-Fi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

Other components not shown in FIG. 6 can be used in one or more embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and does not otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

In the subject specification, terms such as “store,” “storage,” “data store,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, non-volatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SL-

DRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after addition to an existing communication network) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, $x=(x_1, x_2, x_3, x_4 \dots x_n)$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence (class)}$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator prefer-

ences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to predetermined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the

art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

Moreover, terms such as “user equipment,” “mobile station,” “mobile,” subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

As used herein, terms such as “data storage,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be

either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

As may also be used herein, the term(s) “operably coupled to,” “coupled to,” and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that

21

have not been described in the subject disclosure. Further, more than or less than all of the features described with respect to an embodiment can also be utilized.

What is claimed is:

1. A method, comprising:
 - discovering, by a processing system including a processor, an access mobility management function (AMF) in another radio technology network responsive to a message received by a network repository function (NRF) from a mobile management entity (MME) in a Long-Term Evolution (LTE) network that a user equipment (UE) has crossed a threshold and entered a coverage area of the another radio technology network, wherein the UE accesses a packet data network (PDN) through the LTE network;
 - transferring, by the processing system, metadata associated with the UE to the AMF responsive to discovering the AMF;
 - handing over, by the processing system, cellular services for the UE to the another radio technology network after the transferring; and
 - providing, by the processing system, the UE with access to the PDN through a user data plane function (UPF) of the another radio technology network after the handing over.
2. The method of claim 1, wherein the AMF provides the cellular services to the coverage area of the another radio technology network, and wherein the another radio technology network is a fifth generation (5G) network.
3. The method of claim 2, further comprising: checking, by the processing system, that the UE is authorized to access the 5G network before the handing over.
4. The method of claim 2, wherein the NRF discovers the AMF.
5. The method of claim 4, wherein the NRF discovers the AMF without using a domain name service (DNS) protocol.
6. The method of claim 5, wherein the message is passed over a first Representational State Transfer (REST) Application Programming Interface (API).
7. The method of claim 6, wherein the MME performs the transferring by passing a second message over an enhanced N26 interface to the AMF using a second REST API.
8. The method of claim 7, wherein the NRF and the AMF are virtual network elements and wherein the metadata comprises status, heartbeat timer, Internet Protocol address, capacity, priority, load, locality, services supported, or a combination thereof.
9. A device, comprising:
 - a processing system including a processor; and
 - a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising:
 - discovering an access mobility management function (AMF) in another radio technology network responsive to a message received by a network repository function (NRF) from a mobile management entity (MME) in a Long-Term Evolution (LTE) network that a user equipment (UE) has crossed a threshold and entered a coverage area of the another radio technology network, wherein the UE accesses a packet data network (PDN) through the LTE network;

22

- transferring metadata associated with the UE to the AMF responsive to discovering the AMF;
- handing over cellular services for the UE to the another radio technology network after the transferring; and
- providing access to the PDN through a user data plane function (UPF) of the another radio technology network after the handing over.
10. The device of claim 9, wherein the AMF provides services to the coverage area of the another radio technology network.
11. The device of claim 9, wherein the NRF discovers the AMF.
12. The device of claim 11, wherein the message is passed over a first Representational State Transfer (REST) Application Programming Interface (API).
13. The device of claim 11, wherein the MME performs the transferring using a second REST API to the AMF over an enhanced N26 interface.
14. The device of claim 11, wherein the NRF discovers the AMF without using a domain name service (DNS) protocol.
15. The device of claim 11, wherein the NRF and the AMF are virtual network elements.
16. The device of claim 9, wherein the processing system comprises a plurality of processors operating in a distributed computing environment.
17. A non-transitory, machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising:
 - discovering an access mobility management function (AMF) in another radio technology network responsive to a message received by a network repository function (NRF) from a mobile management entity (MME) in a Long-Term Evolution (LTE) network that a user equipment (UE) has crossed a threshold and entered a coverage area of the another radio technology network, wherein the UE accesses a packet data network (PDN) through the LTE network;
 - transferring metadata associated with the UE to the AMF responsive to discovering the AMF;
 - handing over cellular services for the UE to the another radio technology network after the transferring; and
 - providing access to the PDN through a user data plane function (UPF) of the another radio technology network after the handing over.
18. The non-transitory, machine-readable medium of claim 17, wherein the processing system comprises a plurality of processors operating in a distributed computing environment.
19. The non-transitory, machine-readable medium of claim 17, wherein the NRF discovers the AMF.
20. The non-transitory, machine-readable medium of claim 19, wherein the message is passed over a first Representational State Transfer (REST) Application Programming Interface (API), and wherein the MME performs the transferring using a second REST API to the AMF over an enhanced N26 interface.

* * * * *